Second-Order Finite Differences Invert the Geometry of Generative Models: A Non-Recoverability Result and an Exact Instrument

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Abstract

Studies of the differential geometry of generative models increasingly measure *second-order* quantities—curvature of the generated manifold and the terms that enter geodesics. We report a concrete, reproducible hazard in how these are computed. On Stable Diffusion, a representative second-order geometric measurement—the rank correlation between the first-order singular value σ of the latent-to-image Jacobian and a second-order curvature ratio ρ along the leading directions—comes out *with opposite sign* depending only on the numerical method: the exact forward-mode estimate gives a consistent strong negative correlation (Spearman -0.73, 5/6 seeds ≤ -0.82), whereas a finite-difference (FD) estimate of the same quantity is sign-unstable and averages -0.18. The first-order Jacobian, by contrast, is recovered by finite differences to 2.4%. We explain this asymmetry: central FD for a *second* derivative is *non-recoverable*—truncation and floating-point cancellation trade off so that no step size beats a $\sqrt{\varepsilon_{\text{mach}}}$ relative-error floor, which in the single precision generators use is large—while FD for a first derivative is well conditioned. On a StyleGAN generator the naive FD curvature estimate misses the exact value by $\geq 45\%$ at *every* step (validated against the exact instrument by the first-order FD, which agrees to 2.4%). We provide the exact constant-memory instrument—composed forward-mode automatic differentiation, generator-agnostic and extending to any order—and survey how the literature computes second-order geometry, finding that the catastrophic explicit-$/\varepsilon^2$ regime is the natural first thing to reach for but is, fortunately, avoided by most published methods. Our contribution is the precise account of when finite differences break second-order generative geometry, the practitioner-facing first/second-order asymmetry, and a drop-in exact replacement.

1 Introduction

Let $G : \mathbb{R}^d \to \mathbb{R}^{3HW}$ be a trained generator. Treating $\{G(z)\}$ as a manifold pulled back through G, a productive line of work studies its geometry: the pullback metric $M_z = J_z^\top J_z$ and its singular vectors (the local latent basis), geodesics between latent codes, and the curvature of the generated manifold (Arvanitidis et al., 2018; Shao et al., 2018; Chen et al., 2018; Park et al., 2023; Saito & Matsubara, 2025). First-order objects (J_z, M_z) are routine to compute. The *second-order* objects —curvature, the second-order pushforward $\partial^2 G/\partial\alpha^2$ along a path, the metric derivative $\partial_z M_z$ that appears in the geodesic ODE—are where numerical practice matters and, we show, can silently flip a result.

Our entry point is an observation, not a theorem (Section 2). A standard second-order measurement on Stable Diffusion returns the *opposite sign* depending only on whether the second derivative is taken by finite differences or by exact forward-mode AD. The exact method finds a clean, seed-consistent relationship; the finite-difference method, applied to the very same quantity, is sign-unstable and washes it out. A practitioner choosing the natural finite-difference route would draw the wrong qualitative conclusion.

The explanation is an asymmetry that is elementary once stated but, in our experience, not widely appreciated in this applied setting (Section 3): a central finite difference is well conditioned for a *first* derivative but *non-recoverable* for a *second* derivative, because for the second derivative truncation error and floating-point cancellation cannot be made small at the same step size, leaving a relative-error floor of order $\sqrt{\varepsilon_{\text{mach}}}$. In the single precision that GANs and diffusion U-Nets run in, that floor is large, and on a strongly nonlinear generator it is larger still (Section 4: $\geq 45\%$ on StyleGAN, at every step).

We make three contributions: (i) the reproducible sign-inversion phenomenon and the first/second-order asymmetry that governs it, as concrete guidance; (ii) an exact, constant-memory instrument for the higher-order pushforward via composed forward-mode AD, which removes the hazard at the cost of one extra forward pass per order; and (iii) a survey of how the field computes second-order geometry, locating where the trap is and is not. We are explicit about scope (Section 7): we do not claim any published result is wrong—indeed we find most published methods route around the catastrophic regime—we characterize a real numerical hazard and supply the fix.

2 The sign-inversion phenomenon

On a Stable-Diffusion h-space pipeline we take a base latent, form the first-order Jacobian J of the (truncated) denoising map along K random h-space probes, and along its top-k right-singular directions u_i measure a second-order curvature ratio

$$\rho_i \;=\; \frac{|\,\partial_\alpha^2 G(z+\alpha u_i)\,|}{|\,\partial_\alpha G(z+\alpha u_i)\,|}\Bigg|_{\alpha=0},$$

the magnitude of the second-order pushforward relative to the first. We then ask a standard question—does curvature co-vary with the first-order singular value σ_i?—via Spearman(σ,ρ), computing the second derivative in ρ two ways: exact composed JVP (Section 5) and a finite difference of the JVP (the natural way a practitioner would estimate it).

Over 6 seeds (top-12 directions each), the two methods disagree in *sign* (Table 1). The exact instrument yields a consistent strong negative correlation (mean -0.73; 5/6 seeds ≤ -0.82). The finite-difference estimate of the *same* ratio is sign-unstable across seeds and averages -0.18: it does not merely add noise, it destroys the relationship. A study relying on the finite-difference route would report "no clear relationship," the opposite of what the geometry actually contains.

seed	0	1	2	3	4	5	**mean**
exact composed JVP	-0.85	-0.83	-0.92	-0.94	-0.03	-0.84	$\mathbf{-0.73}$
FD-of-JVP(ε=0.05)	$+0.18$	-0.55	-0.78	-0.40	$+0.55$	-0.07	$\mathbf{-0.18}$

Table 1: Spearman(σ,ρ) on Stable Diffusion, per seed. The exact second-order estimate is consistently strongly negative; the finite-difference estimate of the identical quantity changes sign across seeds and averages near zero. (ρ is a measurement we define to expose the method dependence; the point is that the two numerical routes disagree on its sign, not the specific value of any prior work.)

3 Why: finite differences are non-recoverable for a second derivative

For smooth scalar f the central second difference is $D_\varepsilon^2 f = \big(f(\alpha+\varepsilon) - 2f(\alpha) + f(\alpha-\varepsilon)\big)/\varepsilon^2$, with

$$D_\varepsilon^2 f - f''(\alpha) = \underbrace{\tfrac{\varepsilon^2}{12} f^{(4)}(\alpha) + O(\varepsilon^4)}_{\text{truncation}} + \underbrace{O\big(\varepsilon_{\text{mach}}\,|f|/\varepsilon^2\big)}_{\text{cancellation}}.$$

The two terms cannot be made small together: minimizing their sum gives an optimal step $\varepsilon^\star \sim \varepsilon_{\text{mach}}^{1/4}$ and a relative-error floor $\sim \varepsilon_{\text{mach}}^{1/2}$. In double precision the floor is $\sim 10^{-8}$; in the single precision generators use ($\varepsilon_{\text{mach}} \approx 10^{-7}$) it is $\sim 3 \times 10^{-4}$ in the benign case and far worse when $|f^{(4)}|$ is large. The floor is a property of the *operation*: no choice of ε removes it.

The asymmetry. The *first* central difference $\big(f(\alpha{+}\varepsilon)-f(\alpha{-}\varepsilon)\big)/2\varepsilon$ has truncation $O(\varepsilon^2)$ and cancellation $O(\varepsilon_{\text{mach}}|f|/\varepsilon)$, giving a much milder floor $\sim \varepsilon_{\text{mach}}^{2/3}$ and a wide usable window. This is why first-order Jacobians are safely finite-differenced while second-order curvature is not—and it is the control we use to validate our instrument (Sections 2, 4): wherever the first-order FD agrees with exact JVP, the pipeline is correct and any second-order disagreement is genuine.

Analytic referee. Let $f(\alpha) = [\,e^{0.7\alpha}, \sin 3\alpha, (\alpha+0.4)^3, \tanh 2\alpha, \alpha^2\cos\alpha\,]$, with closed-form derivatives. At $\alpha = 0.3$ in double precision (Table 2) the central-FD second-order relative error is U-shaped, bottoming at 8.6×10^{-9} near $\varepsilon = 10^{-4}$ and diverging to 0.29 by $\varepsilon = 10^{-8}$; the composed JVP attains 2.5×10^{-17}—FD's best is 3.5×10^{8} times worse.

ε	central FD 2nd-order rel. err.	
10^{-1}	7.9×10^{-3}	(truncation)
10^{-2}	8.0×10^{-5}	
10^{-4}	8.6×10^{-9}	(best)
10^{-6}	3.0×10^{-5}	
10^{-8}	2.9×10^{-1}	(cancellation)
composed JVP	2.5×10^{-17}	(exact)

Table 2: Second-order accuracy on the analytic toy at $\alpha = 0.3$, double precision. Finite differences never approach the exact composed JVP.

4 Quantification on a real generator

The toy fixes ideas; the floor is worse in practice. On a StyleGAN-bedroom generator (256×256, fp32) we form the naive FD curvature estimate a practitioner would write, $D^2_\delta G = \big(G(z{+}\delta u) - 2G(z) + G(z{-}\delta u)\big)/\delta^2$, and the exact composed JVP $\partial^2_\alpha G(z{+}\alpha u)$, over 6 directions (3 attributes, 3 random) × 16 samples, sweeping δ (Table 3). The exact curvature is a definite, step-free number (mean 1.17). The FD estimate has a relative-error floor of 45.2% at its best step ($\delta = 3$; bracketed both sides), is 267% at $\delta{\sim}0.3$, and explodes as $\delta \to 0$ (2.0×10^{6}% at $\delta = 10^{-3}$): *no step recovers the curvature.* The control: the *first*-order central difference of the same pipeline agrees with the exact first-order JVP to 2.4%, so the instrument and the pipeline are correct and the second-order blow-up is the non-recoverability of Section 3, not a bug.

step δ	$\lvert D^2_\delta G\rvert$	2nd-order rel. err.	1st-order rel. err.
8	7.1×10^{-3}	74.5%	63.8%
3	2.7×10^{-2}	**45.2**% (floor)	44.1%
1	9.9×10^{-2}	106.4%	26.2%
0.3	3.7×10^{-1}	266.9%	12.5%
0.1	1.06	622.8%	5.5%
0.02	4.05	6.8×10^{3}%	**2.4**% (1st-order min)
10^{-3}	4.09×10^{2}	2.0×10^{6}%	139%
composed JVP	1.17	exact (step-free)	—

Table 3: StyleGAN-bedroom, fp32. The naive FD second difference never gets within 45% of the exact curvature at any step; the first-order FD of the same code agrees to 2.4%, validating the instrument.

5 The exact instrument: composed forward-mode AD

Forward-mode AD evaluates, for a map f and tangent v, the Jacobian–vector product $\partial f(x)[v]$ alongside $f(x)$ in one pass without forming the Jacobian. With $f = G$, $x = \gamma(\alpha)$, $v = u$ this gives the exact first-order pushforward. The second-order term is the directional derivative of the first, so composing forward mode with itself, $\partial_\alpha^2 G(\gamma(\alpha)) = \partial_\alpha\big[\partial G(\gamma(\alpha))[u]\big][u]$, is exact and is evaluated by nesting two JVP calls (`torch.func.jvp` of a function that returns a JVP). No Jacobian is stored, so memory stays at one forward pass; the same nesting yields any order in a constant number of passes. The instrument is generator-agnostic (we ran it unmodified on StyleGAN1/2 and a latent-diffusion U-Net) and, unlike reverse-mode Hessian materialization (memory $O(3HW)$), its cost is independent of output dimension.

6 Survey: where the trap is, and is not

Table 4 classifies representative latent-geometry papers by how they obtain second-order/geodesic quantities. The explicit $/\varepsilon^2$ second-order finite difference—the catastrophic regime of Section 3—is the natural first thing to reach for, but most published methods avoid it: they keep the metric first-order, replace the geodesic ODE with a discretized energy whose *exact* gradient is a well-conditioned discrete Laplacian (not a $/\varepsilon^2$ curvature), use analytic Christoffel symbols, or differentiate the energy with autodiff (Yang et al., 2018); first-order constructions (Park et al., 2023; Chen et al., 2018; Kwon et al., 2023) never touch a second derivative. The genuinely exposed practice is first-order FD of a network derivative (e.g. a score-Jacobian–vector product), which is only mildly inaccurate, and—most relevant here—the direct $/\varepsilon^2$ curvature estimate that a practitioner adds on top of an existing pipeline, as in our Section 2 measurement. The contribution of the survey is to draw this line precisely, so that a reader knows which side they are on.

Paper	2nd-order quantity	how obtained
Shao et al. (2018)	discretized geodesic energy	exact discrete-energy gradient (Laplacian), coarse T; *no* $/\varepsilon^2$
Saito & Matsubara (2025)	geodesic (energy)	first-order FD score-JVP $(s(x+hv)-s(x))/h$
Arvanitidis et al. (2018)	geodesic ODE (Christoffel)	analytic ∂M, numeric ODE integ.
Yang et al. (2018)	geodesic energy	autodiff
Park et al. (2023)	none (first-order basis)	autodiff power iteration
Chen et al. (2018)	none (first-order metric)	—
Kwon et al. (2023)	none (first-order)	CLIP-grad + MLP

Table 4: How representative latent-geometry papers obtain second-order quantities. The explicit $/\varepsilon^2$ second-order finite difference (Sections 3, 4) is largely avoided in published work; it is nonetheless the natural choice for a practitioner adding a curvature measurement, which is where the hazard bites.

Guidance. The practical rule is short. *First*-order quantities (Jacobians, pullback metrics, first-order pushforwards) may be finite-differenced; choose a step near $\varepsilon_{\text{mach}}^{1/3}$ and expect a few digits. *Second or higher*-order quantities (curvature, the second-order pushforward, the metric derivative / Christoffel symbols, anything with a $/\varepsilon^2$) should *not* be finite-differenced: in single precision no step gives even one reliable digit on a deep generator (Section 4), and the error can flip the sign of a downstream conclusion (Section 2). Use composed forward-mode AD instead —it is exact, costs one extra forward pass per order, and adds no memory. A free diagnostic comes with it: compare the first-order finite difference to the first-order JVP; if they disagree, the pipeline is wrong, and if they agree, any second-order disagreement is real, not numerical.

7 Scope, limitations, and what we do not claim

This is a numerical-methods contribution with a survey and demonstrations. **(1)** We do *not* claim any surveyed paper reached a false conclusion; on the contrary, the survey shows most route around the catastrophic regime. The sign inversion of Section 2 is on a quantity *we* define to expose method dependence, not a number from prior work. **(2)** The non-recoverability result is specifically about *second*-order finite differences; first-order finite differences (e.g. score-JVPs) are well conditioned, and we say so. **(3)** The instrument is a thin, exact wrapper around forward-mode AD; its value is in routine exact higher-order pushforward and in the precise account of when the finite-difference alternative fails, not in algorithmic novelty. **(4)** We do not propose a downstream task win; a companion analysis found that curvature, after controlling for edit magnitude, is not an actionable edit-risk signal, and we do not relitigate that here.

8 Reproducibility

We release the exact-instrument module, the analytic referee, and the StyleGAN and Stable-Diffusion scripts with fixed seeds and the emitted `metrics.json` for every table. All experiments run on a single 8 GB GPU.

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